Lonely Runner Conjecture

Audited research report and exact obstruction to a proposed universal-grid extension

Piatra Institute research artifact
Status date: 22 July 2026
Arithmetic: exact rational arithmetic throughout

1. Verdict

The main Lonely Runner Conjecture was **not completely resolved** in this investigation. I found neither:

1. a proof valid for every number of runners; nor
2. an exact speed tuple that violates the conjecture.

Accordingly, this report does not claim a solution. It records the strongest result that survived exact checking and adversarial review:

> **The universal-denominator conjecture proposed as Conjecture 7.1 in Sungkawichai and Trakulthongchai (2026) is false as written, and fails in two robust ways.**

The report also gives:

- a complete symbolic proof of that statement;
- an exact critical-time lemma for concrete speed tuples;
- a corrected quantitative grid theorem;
- two machine-checkable certificates in the first currently unverified dimension, $k=13$ relative speeds;
- exact Python verification code using only the standard library;
- the precise remaining gap to a complete resolution of the Lonely Runner Conjecture.

This is a rigorous correction to a proposed extension mechanism, but it is **not** a resolution of the original conjecture. I did not locate this specific obstruction in targeted public searches; that is not a certification of literature priority, and this report makes no novelty claim beyond the derivations it contains.

2. Problem and conventions

For a real number x, write $||x||$ for its distance to the nearest integer. For a tuple of k nonzero integer relative speeds

$v = (v_1,\dots,v_k)$,

set

F_v(t) = min_i ||t v_i||,
kappa(v) = max_{t in R/Z} F_v(t).

The stationary-runner form of the Lonely Runner Conjecture is

kappa(v) >= 1/(k+1)

for every admissible tuple. The threshold uses k+1 because k nonzero relative speeds arise after fixing one runner among k+1 total runners.

Negative speeds cause no change because ||-x||=||x||. A common greatest common divisor causes no change after rescaling time. If two signed speeds have the same absolute value, they impose the same constraint; deleting the duplicate only strengthens the threshold available from the lower-dimensional conjecture.

As of 22 July 2026, the strongest publicly available verification proves the conjecture for k <= 12, corresponding to at most 13 total runners. The first unverified stationary case is k=13, or 14 total runners.

3. The proposed universal-grid conjecture

Sungkawichai and Trakulthongchai define a positive integer speed tuple to be **tight** when the strict inequalities

||t v_i|| > 1/(k+1) for every i

cannot hold simultaneously. Their Conjecture 7.1 states, in substance:

> For each fixed k, there is a constant D such that, for every integer d >= D, every coprime non-tight positive integer k-tuple has a witness in (1/d)Z.

The following theorem disproves this statement for every k >= 2. The tuple is allowed to depend on d: the conjecture quantifies over **every** non-tight coprime tuple after fixing the denominator, so one counterexample tuple for each large d is decisive.

4. Main theorem established here: failure for every sufficiently large denominator

Theorem 4.1 - diagonal grid obstruction

Fix an integer k >= 2, put q=k+1, and let

C_k = k(k+1)(k-1)/2.

For every integer d > C_k, the tuple

V_{k,d} = (1,2,...,k-1,d)

has all of the following properties:

1. its entries are positive and distinct;

2. $\gcd(V_{k,d})=1$;
3. it is non-tight, in fact $\kappa(V_{k,d}) > 1/q$;
4. it has no witness in $(1/d)\mathbb{Z}$.

Consequently, no constant D of the form proposed in Conjecture 7.1 can exist.

Proof

The first two statements are immediate because the tuple contains 1 and $d>k-1$.

For the grid obstruction, take any grid time $t=a/d$. The last coordinate satisfies

$$\|d\,t\| = \|a\| = 0 < 1/q.$$

Therefore no point of $(1/d)\mathbb{Z}$ is a witness.

It remains to prove that the tuple is nevertheless non-tight. At

$$t_0=1/k,$$

for every $i=1,\dots,k-1$,

$$\|i\,t_0\| = \min(i/k,1-i/k) \geq 1/k.$$

The margin above the target threshold is

$$1/k - 1/(k+1) = 1/[k(k+1)].$$

Choose a midpoint of a d-grid cell,

$$t_*=(2m+1)/(2d),$$

nearest to $1/k$. Such a midpoint satisfies

$$|t_*-1/k| \leq 1/(2d).$$

The distance-to-nearest-integer function is 1-Lipschitz, hence for $1\leq i\leq k-1$,

$$\|i\,t_*\| \geq \|i/k\| - i|t_*-1/k|$$

$$\geq 1/k - (k-1)/(2d).$$

Because $d>C_k$,

$$1/k - (k-1)/(2d) > 1/(k+1).$$

For the last speed,

$$\|d\,t_*\| = \|(2m+1)/2\| = 1/2 > 1/(k+1).$$

Thus every inequality is strict at t_*, so $V_{k,d}$ is non-tight. This completes the proof. □

Quantitative form

The proof gives the explicit lower bound

$\kappa(V_{k,d}) \geq 1/k - (k-1)/(2d) > 1/(k+1)$.

For the concrete k=13, d=2000 certificate supplied with this report, the stronger identity

$\kappa(1,2,\ldots,12,2000)=1/13$

also has a short symbolic proof. The prefix (1,...,12) has maximum loneliness 1/13 by Lemma 5.1, so adjoining another coordinate cannot increase the maximum. At t=1/13, the added speed is congruent to 11 mod 13, and every coordinate has distance at least 1/13; hence equality holds. There are no witnesses on the 1/2000 grid, although the required threshold is only 1/14.

5. Robust failure even when every speed is a unit modulo the grid denominator

A natural attempted repair is to prohibit speeds divisible by the grid denominator. That does not repair the conjecture.

Lemma 5.1 - exact witnesses for the canonical tight tuple

For q=k+1,

$\min_{1\leq i\leq k} \|i t\| \geq 1/q$

holds if and only if

$t = s/q \bmod 1$

for some s with gcd(s,q)=1.

Proof

Consider the q points

$0,t,2t,\ldots,kt$ in $\mathbb{R}/\mathbb{Z}$.

The distance between any two is $\|(i-j)t\|$, where $1\leq|i-j|\leq k$; by hypothesis every pair is separated by at least 1/q. The q cyclic gaps between the sorted points therefore each have length at least 1/q. Their lengths sum to 1, so every gap is exactly 1/q. The points form a regular q-gon. Hence t=s/q mod 1; the points are distinct precisely when gcd(s,q)=1. The converse follows immediately from the nonzero residues modulo q. □

Theorem 5.2 - congruence-class obstruction

Fix k>=2 and put q=k+1. For every sufficiently large integer d coprime to q, the tuple

$W_{k,d}=(1,2,\ldots,k-1,d+k)$

is coprime and non-tight but has no witness in $(1/d)\mathbb{Z}$.

If $d>k$ is prime, every individual speed in $W_{k,d}$ is coprime to d.

Proof

At every grid time $t=a/d$, the tuple is congruent modulo d to $(1,2,\dots,k)$, because

$$(d+k)a/d = a + ka/d.$$

Thus a grid witness for $W_{k,d}$ would also be a grid witness for $(1,\dots,k)$. By Lemma 5.1 it would have to satisfy

$$a/d=s/q \mod 1$$

with $\gcd(s,q)=1$. Since $\gcd(d,q)=1$, no nonzero such time belongs to both grids. Therefore no grid witness exists.

Non-tightness follows from exactly the midpoint argument in Theorem 4.1, now using the last speed $d+k$; it is enough that

$$d+k > C_k.$$

If $d>k$ is prime, the residues $1,\dots,k$ are all nonzero modulo d, so every speed is individually coprime to d. □

For the supplied robust certificate,

$$k=13,\ d=2003,\ W=(1,2,\dots,12,2016).$$

Here 2003 is prime and every speed is coprime to 2003. Moreover, 2016 is congruent to 1 modulo 13, so the same prefix upper bound and the witness $t=1/13$ prove symbolically that the exact maximum loneliness is $1/13$. The $1/2003$ grid contains no witness.

6. Why the failure is structural

The fixed grid sees only the residue class of each speed modulo its denominator. A high-frequency lift can preserve all grid data while changing the continuous-time geometry from tight to non-tight.

Lemma 6.1 - congruence-blind lifting

Fix a threshold $\alpha<1/2$, a denominator d, and a tuple $A=(a_1,\dots,a_k)$ having no witness on the $1/d$ grid. Suppose that, after deleting coordinate j, there is an open interval I on which every remaining coordinate is strictly above α.

Replace a_j by

$$a_j+m\,d.$$

For all sufficiently large positive integers m, the new tuple:

1. has exactly the same behavior as A on the $1/d$ grid;
2. has a strict continuous-time witness in I.

Proof

Grid behavior is unchanged because the replacement is congruent to a_j modulo d.

For frequency w=a_j+md, the bad set ||w t||<=alpha is a union of intervals, each of length 2alpha/w. Once w is large enough that 2alpha/w<|I|, the whole interval I cannot be contained in one bad component. Hence some t in I satisfies ||w t||>alpha; the other coordinates were already strict throughout I. □

This lemma explains why a denominator independent of the tuple cannot distinguish tight residue data from non-tight high-frequency lifts.

7. Exact critical-time lemma

The checker uses the following finite characterization.

Lemma 7.1 - a maximum occurs at a pair-sum denominator

Let v_1,...,v_k be nonzero integers and

F(t)=min_i ||v_i t||.

There is a maximizing time of the form

t=a/(|v_i|+|v_j|)

for some indices i,j, allowing i=j, and some integer a.

Proof

Each function ||v_i t|| is a triangular wave: continuous and piecewise affine with slopes +|v_i| and -|v_i|. Hence F, the lower envelope, is continuous and piecewise affine.

Choose a maximizer at an endpoint if the maximizing set contains an interval. If an active triangular wave has a breakpoint there, its denominator divides 2|v_i|=|v_i|+|v_i|.

Otherwise, at a local maximum of the lower envelope, active affine branches of both slope signs must meet. Equality between a branch of slope +|v_i| and one of slope -|v_j| gives

(|v_i|+|v_j|)t equal to an integer.

This proves the claim. □

Corollary 7.2 - exact rationality and slack quantization

Let M=max_i |v_i|, q=k+1, and suppose

kappa(v)>1/q.

Then for some N<=2M,

kappa(v)=m/N

with integer m, and therefore

kappa(v)-1/q >= 1/(qN) >= 1/(2qM).

This gives a completely explicit separation between a non-tight integer tuple and the boundary once its speed scale is fixed.

8. Corrected quantitative grid theorem

The failed conjecture becomes true after the denominator is allowed to depend on the speed scale.

Theorem 8.1 - speed-bounded universal grid

Let v be a non-tight integer k-tuple, let M=max_i |v_i|, and put q=k+1. Then every integer grid denominator

d >= q M^2

contains a witness for v.

More sharply, if a maximizing time in Lemma 7.1 has denominator N, then

d >= q M N/2

is sufficient.

Proof

Take a maximizing time t_0=a/N. By Corollary 7.2,

kappa(v)-1/q >= 1/(qN).

Choose b/d within circular distance 1/(2d) of t_0. Every function ||v_i t|| is |v_i|-Lipschitz, so their minimum is M-Lipschitz. Therefore

F(b/d) >= kappa(v)-M/(2d).

If d>=qMN/2, then M/(2d)<=1/(qN), and hence F(b/d)>=1/q. Since N<=2M, the simpler condition d>=qM^2 suffices. □

Near-necessity of speed dependence

Theorem 4.1 has maximum speed M=d and no witness on the 1/d grid. Thus no denominator threshold independent of M can work, and even a general sublinear-in-M threshold is impossible. The exact optimal growth rate between the linear obstruction and the quadratic upper bound remains open.

9. Audited routes toward the main conjecture

The following routes were pursued or checked. None produced a complete all-k proof or an exact counterexample.

9.1 Fixed universal grids

Outcome: blocked decisively by Theorems 4.1 and 5.2.

A fixed denominator cannot see high-frequency changes by multiples of that denominator. This is not a numerical issue; it is an exact congruence obstruction.

9.2 Exact finite critical times

Outcome: valid and implemented, but tuple-dependent.

Lemma 7.1 makes every concrete tuple exactly decidable. It does not provide a bound independent of the input speeds or of k, so it does not by itself meet the complete-resolution standard.

9.3 Prefix induction with slack

Assuming the conjecture for k-1 speeds, if the sorted tuple satisfies

$v_k > k v_{k-1}$,

a witness for the first k-1 speeds at threshold 1/k gives an interval on which they remain above 1/(k+1); the last runner oscillates fast enough that its safe set intersects that interval. This is a valid lacunary induction step, but it leaves the medium-growth regime.

At the opposite extreme, if

$v_k \le k v_1$,

then $t=1/(v_1+v_k)$ is already a witness. The unresolved regime lies between these easy geometric extremes and includes highly structured tuples.

9.4 Shifted or phase-perturbed induction

Outcome: unsafe as a general mechanism.

Perturbing a rational base time turns the remaining constraints into shifted simultaneous approximation conditions. The shifted Lonely Runner Conjecture is not equivalent to the ordinary conjecture and now has explicit counterexamples, so a proof step that silently invokes the shifted statement is circular or false.

9.5 Fourier and measure arguments

Outcome: rigorous lower bounds remain below the target.

The best current asymptotic Riesz-product bound has the form $1/(2k)+1/k^{(5/3+o(1))}$; the target 1/(k+1) is asymptotically 1/k, so a factor-two leading-order gap remains. For a tight tuple the strict-witness set is empty, so a positive-measure argument cannot by itself establish the inclusive boundary equality.

9.6 Bounded computation in the first unknown case

Exploratory exact and mixed-integer searches did not produce a $k=13$ counterexample in the bounded regions tested. Such a non-result is not evidence of the conjecture and is not used in any theorem in this report. A bounded search without a proved complete bound cannot meet the prompt's resolution standard.

10. Exact remaining gap

A complete solution still requires exactly one of the following:

1. a uniform proof that kappa(v)>=1/(k+1) for every k and every admissible tuple; or
2. a finite exact tuple with kappa(v)<1/(k+1), together with a symbolic full-circle forbidden-interval cover.

The results above do neither. They establish that one proposed route to a uniform proof is impossible as stated and replace it with a correct speed-dependent theorem.

11. Stationary and moving-runner translation

Given k+1 runners with distinct speeds w_0,...,w_k, fix runner r and subtract its speed. The k relative speeds

u_i=w_i-w_r, i != r,

are distinct and nonzero. A stationary-form witness satisfies

||t(w_i-w_r)|| >= 1/(k+1)

for every other runner, exactly saying that runner r is lonely at time t. Applying the stationary theorem separately for each choice of r gives the usual statement that every runner is lonely at some possibly different time.

Conversely, append a stationary runner of speed 0 to any stationary-form tuple. The boundary is preserved exactly because the conjecture uses >=, not >.

12. Reproduction

The package contains:

- lrc_exact.py: exact checker and critical-time evaluator;
- lrc_grid_obstruction_certificate.json: two exact k=13 certificates;
- verify_certificate.py: independent command-line wrapper;
- RESEARCH_LOG.md: approach registry and blockers;
- SHA256SUMS: integrity hashes.

Run:

```
python lrc_exact.py --self-test
python verify_certificate.py lrc_grid_obstruction_certificate.json
```

To reproduce the exact diagonal example:

```
python lrc_exact.py \
  --speeds 1 2 3 4 5 6 7 8 9 10 11 12 2000 \
  --grid 2000 --maximum
```

To reproduce the unit-modulo-grid example:

```
python lrc_exact.py \
  --speeds 1 2 3 4 5 6 7 8 9 10 11 12 2016 \
  --grid 2003 --maximum
```

Expected in both examples:

- threshold 1/14;
- grid witness count 0;
- exact maximum loneliness 1/13.

13. References

1. T. Sungkawichai and T. Trakulthongchai, *Eleven, twelve, and thirteen lonely runners*, arXiv:2604.23906, 2026.
2. R. D. Malikiosis, F. Santos, and M. Schymura, *Linearly-exponential checking is enough for the Lonely Runner Conjecture and some of its variants*, arXiv:2411.06903, revised 2025.
3. V. Giri and N. Kravitz, *The structure of Lonely Runner spectra*, Mathematical Proceedings of the Cambridge Philosophical Society 180 (2026), 343-361.
4. B. Bedert, *Riesz products and the Lonely Runner Conjecture: A wider gap of loneliness*, arXiv:2511.16636, 2025.
5. M. Blanco, F. Criado, and F. Santos, *Coloopless and cosimple zonotopes, and the Lonely Runner Conjectures*, arXiv:2603.24784, 2026.
6. G. Perarnau and O. Serra, *The Lonely Runner Conjecture turns 60*, arXiv:2409.20160, 2024.

14. Final takeaway

The original Lonely Runner Conjecture remains open beyond 13 total runners. The main rigorously verified advance here is that the 2026 universal-denominator proposal is false in every dimension k>=2, including under the natural repair that all speeds be coprime to a prime grid denominator. The correct replacement must depend on the tuple's speed scale or an equivalent quantitative compactness parameter; d >= (k+1)M^2 is one explicit sufficient bound.